List of Publications

Tayyab Tahir
Tallinn, Estonia | e.tayyab1024@gmail.com | Google Scholar | ORCID | tayyab-tahir1.github.io
168 citations, h-index 6, i10-index 4 (Google Scholar, October 2026)

Journal articles

[1] M. Irfan, **T. Tahir**, S. Deilami, S. Huang, B. P. Veettil, "Novel artificial intelligence driven model-based control framework for solar–electric vehicle home energy optimization," *Engineering Applications of Artificial Intelligence*, vol. 177, Art. no. 114868, 2026. doi:10.1016/j.engappai.2026.114868

[2] M. Irfan, **T. Tahir**, S. Deilami, S. Huang, B. P. Veettil, "Novel control-based design optimization of smart energy distribution and management in vehicle-to-grid integrated microgrid," *Applied Energy*, vol. 402, Art. no. 127047, 2026. doi:10.1016/j.apenergy.2025.127047

[3] M. Irfan, **T. Tahir**, S. Huang, S. Deilami, B. P. Veettil, "Optimizing residential energy costs: A novel machine learning approach for solar PV and EV integration through heuristic price signal dispatch," *IEEE Transactions on Industry Applications*, vol. 61, no. 3, pp. 4684–4694, 2025. doi:10.1109/TIA.2025.3537072

[4] M. Irfan, S. Deilami, S. Huang, **T. Tahir**, B. P. Veettil, "Optimizing load frequency control in microgrid with vehicle-to-grid integration in Australia: Based on an enhanced control approach," *Applied Energy*, vol. 366, Art. no. 123317, 2024. doi:10.1016/j.apenergy.2024.123317

[5] **T. Tahir**, A. Khalid, J. Arshad, A. Haider, I. Rasheed, A. U. Rehman, S. Hussen, "Implementation of an IoT-based solar-powered smart lawn mower," *Wireless Communications and Mobile Computing*, vol. 2022, no. 1, Art. no. 1971902, 2022. doi:10.1155/2022/1971902

Conference papers

[6] M. Irfan, **T. Tahir**, S. Deilami, S. Huang, B. P. Veettil, "A novel predictive strategy for daily energy cost reduction in solar 'PV + EV' system with V2G and V2H modes," in *Proc. IEEE International Conference on Energy Technologies for Future Grids (ETFG)*, 2025, pp. 1–6. doi:10.1109/ETFG61999.2025.11401228

[7] M. Irfan, **T. Tahir**, S. Deilami, S. Huang, B. P. Veettil, "A novel data-driven optimization for cost-effective home energy management with PV–EV integration," in *Proc. IEEE 34th Australasian Universities Power Engineering Conference (AUPEC)*, 2024, pp. 1–6. doi:10.1109/AUPEC62273.2024.10807596

[8] Z. M. Tahir, **T. Tahir**, T. Arfan, A. Asim, M. Fahad, S. Rauf, "Design and implementation of prepaid energy meter with home automation and peak load management using Android application," *Engineering Proceedings* (IEEC 2023), vol. 46, no. 1, Art. no. 23, 2023. doi:10.3390/engproc2023046023

[9] A. Khalid, **T. Tahir**, A. Haider, N. M. Ahmed, M. Hamid, S. Attari, "Autonomous vehicle based on lane detection through computer vision using Raspberry Pi," in *Proc. International Conference on IT and Industrial Technologies (ICIT)*, 2022, pp. 1–7. doi:10.1109/ICIT56493.2022.9989134

My contribution

Journal articles [1]–[3] and conference papers [6]–[7]: I developed the deep-RL and control layer and the EV battery model, put the battery constraints into the reward, ran the training on a computing cluster, and drafted the corresponding methodology sections and reviewer responses. Journal article [4]: Software (Simulink plant model, PSO tuning, ANN).

Manuscript in preparation

- Uncertainty-aware transformer for rover wheel-slip estimation from multi-rate proprioceptive streams under scarce labels and sensor loss (TalTech Centre for Biorobotics), for submission to *Engineering Applications of Artificial Intelligence*.

Thesis

- **T. Tahir**, “Modeling and control of smart integrated port energy systems: A data-driven approach,” MSc thesis, Department of Marine Technology, NTNU, Trondheim, 2025. Supervisor: Prof. Mehdi Zadeh.